

Phase 2 — Active Directory and Windows Server

Harrington Capital plc | Infrastructure Engineer Simulation

Objective

Build and document Harrington Capital's Active Directory estate on HC-DC01. The previous engineer (Callum Reid) left no documentation. You are starting from a bare Windows Server 2022 installation.

By the end of this phase you will have: - AD DS installed and HC-DC01 promoted to Domain Controller - Domain: `harrington.local` - Full OU structure documented and built - 10 user accounts created across departments - DNS verified and resolving - Everything documented and committed to GitHub

BUILD

Step 1 — Install AD DS (run as Administrator on HC-DC01)

Open PowerShell 7 as Administrator and run:

```
cd C:\path\to\harrington-infrastructure-operations\powershell
.\01-ad-setup.ps1
```

This script: - Installs the AD DS Windows Feature - Promotes HC-DC01 to Domain Controller for `harrington.local` - Installs DNS Server role automatically - **Restarts the server** — this is expected

Wait for the server to restart (approximately 3-5 minutes).

Step 2 — Create OU Structure and Users (after restart)

Log in to HC-DC01 as `HARRINGTON\Administrator`.

Open PowerShell 7 as Administrator:

```
cd C:\path\to\harrington-infrastructure-operations\powershell  
.\02-ad-ou-users.ps1
```

This script: - Creates the full OU structure under `harrington.local` - Creates 10 user accounts across 5 departments - Runs verification and prints output

Copy the entire VERIFICATION OUTPUT section — you will paste it into your submission.

Step 3 — Verify DNS resolution

On HC-DC01, run:

```
Resolve-DnsName "harrington.local"  
Resolve-DnsName "HC-DC01.harrington.local"  
nslookup harrington.local
```

On another VM (HC-APP01), set DNS to point to HC-DC01's IP, then:

```
nslookup harrington.local <HC-DC01-IP>  
dig harrington.local @<HC-DC01-IP>
```

Step 4 — Take verification screenshots

Open **Active Directory Users and Computers** (dsa.msc): - Expand `harrington.local` - Expand the full OU tree - Screenshot showing all OUs visible

Open **DNS Manager** (dnsmgmt.msc): - Expand Forward Lookup Zones - Screenshot showing `harrington.local` zone

Step 5 — Document the OU structure

Open `/architecture/hc-ad-structure.drawio` (create it in Draw.io).

Draw the OU hierarchy:

```
harrington.local
└─ Harrington Capital (OU)
    └─ Users (OU)
        └─ Technology
        └─ Finance
        └─ Investments
        └─ Risk and Compliance
        └─ Operations
    └─ Computers (OU)
        └─ Workstations
        └─ Servers
    └─ Groups (OU)
    └─ Service Accounts (OU)
```

Export as PNG. Commit both files.

Step 6 — Commit your scripts and diagram

```
git add architecture/hc-ad-structure.drawio architecture/hc-ad-structure.png
git commit -m "docs(architecture): AD OU structure diagram - Phase 2"
git push origin main
```

VERIFY

Run these PowerShell checks on HC-DC01 and copy the output:

```
# Domain info
Get-ADDomain | Select-Object Name, DomainMode, PDCEmulator

# All OUs
Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName | Format-Table -AutoSize

# User count by department
$base = "OU=Users,OU=Harrington Capital,DC=harrington,DC=local"
foreach ($dept in @("Technology","Finance","Investments","Risk and Compliance","Operations")) {
    $count = (Get-ADUser -Filter * -SearchBase "OU=$dept,$base").Count
    Write-Host "$dept : $count users"
}

# Total users
(Get-ADUser -Filter * -SearchBase $base).Count

# DNS test
Resolve-DnsName "harrington.local"

# Services running
Get-Service ADWS, DNS, KDC, Netlogon | Select-Object Name, Status
```

All services must show `Running`. Total users must be **10**.

SUBMIT

Your Phase 2 submission must include:

#	Evidence	How to get it
1	PowerShell VERIFICATION OUTPUT	Copy from script output after running <code>02-ad-ou-users.ps1</code>
2	AD Users and Computers screenshot	dsa.msc — full OU tree visible
3	DNS Manager screenshot	dnsmgmt.msc — harrington.local zone
4	OU structure diagram	<code>/architecture/hc-ad-structure.png</code>
5	GitHub commit URL	Commit adding diagram to repo

Anti-fake check: Your submission must include a user count per OU. The correct counts are: Technology 4, Finance 2, Investments 1, Risk and Compliance 2, Operations 1.

Troubleshooting

Error	Fix
<code>Install-ADDSTForest: The specified domain already exists</code>	Domain was already created — skip to Step 2
<code>Access denied</code>	Ensure PowerShell is running as Administrator
<code>Cannot connect to domain</code>	Check HC-DC01 IP and firewall — port 389 (LDAP) must be open
DNS not resolving from HC-APP01	Set HC-APP01 DNS to HC-DC01 IP: <code>sudo nano /etc/resolv.conf</code>
Users not visible in dsa.msc	Close and reopen Active Directory Users and Computers